Projectile Motion with Air Resistance

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August 18, 2026

1 Physical Model

1.1 Motivation

When we consider a projectile launched from ground level with an initial velocity v_0 at an angle θ above the horizontal, the motion can be described by Newton's second law. If we ask the question of which angle provides the maximum range, in the absence of air resistance, the answer is famously 45°. However this is not the case if we include air resistance.

1.2 The Drag Force

1.3 The Drag Force

When air resistance is included, the projectile is subject to a drag force $\mathbf{F}_r$ that always acts in the opposite direction of the velocity $\mathbf{v}$. Throughout this document, bold symbols denote vectors (e.g. $\mathbf{v}$), while the same symbol in italics denotes its magnitude (e.g. $v = |\mathbf{v}|$).

The form of the drag force depends on the Reynolds number,

$$\mathrm{Re} = \frac{Dv\rho}{\eta}, \tag{1}$$

where D is the projectile's size, ρ the density of air, and η its viscosity.

Re is the ratio of inertial to viscous forces, inside of a moving fluid. If we have a low Re, viscous forces dominate and the drag force is linear,

$$F_r \propto v \qquad \text{(linear, or Stokes, drag)}, \tag{2}$$

which is a regime typical of small objects moving slowly through a viscous fluid, such as a droplet. This regime holds for $\mathrm{Re} \lesssim 1$; for example, a 10 μm fog droplet settling in air at roughly 2.7 cm/s has $\mathrm{Re} \approx 0.016$ [1]. Writing this as a vector,

$$\mathbf{F}_r = -b\,\mathbf{v}, \qquad b = 3\pi\eta D, \tag{3}$$

where b is the linear drag coefficient.

On the other hand, if we have a high Re, inertial forces dominate and the drag force is quadratic,

$$F_r \propto v^2 \qquad \text{(quadratic, or Newtonian, drag)}, \tag{4}$$

which is a regime typical of bigger objects moving through air, such as a thrown ball or a fired projectile. This regime holds for speeds between approximately 0.25 m/s and 53 m/s, corresponding to $10^3 < \mathrm{Re} < 2 \times 10^5$ [2].

Writing the quadratic drag force as a magnitude times a unit vector opposing $\mathbf{v}$,

$$\mathbf{F}_r = -C\,v^2\,\frac{\mathbf{v}}{v} = -C\,v\,\mathbf{v}, \qquad v = |\mathbf{v}| = \sqrt{v_x^2 + v_y^2}, \tag{5}$$

where C is a drag coefficient, assumed constant.

2 System of Differential Equations

3 Numerical Integration

4 Stopping Condition

5 How this turns into Code

References

[1] M. V. Lubarda, V. A. Lubarda, *A review of the analysis of wind-influenced projectile motion in the presence of linear and nonlinear drag force*, Archive of Applied Mechanics **92** (2022) 1997–2017.

[2] P. S. Chudinov, V. A. Eltyshev, Y. A. Barykin, *Simple analytical description of projectile motion in a medium with quadratic drag force*, Latin American Journal of Physics Education **7** (2013).